

Lesson 10: Quadratic Functions

■ Rubric Criteria for Quadratic Functions

- **Conceptual Understanding (Graphing)** (25%): Identifies key features including vertex, axis of symmetry, x -intercepts, and y -intercepts.
- **Algebraic Skills (Solving)** (50%): Demonstrates proficiency in solving quadratic equations using multiple methods: factoring, completing the square, and the quadratic formula.
- **Application/Modeling** (15%): Translates real-life scenarios into quadratic functions to solve problems involving maxima/minima (e.g., maximum height, profit).
- **Representations** (10%): Converts between standard form ($y = ax^2 + bx + c$), vertex form ($y = a(x - h)^2 + k$), and factored form ($y = a(x - r)(x - s)$) to identify properties.